Simulated Annealing on a Discrete Grid

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1 Introduction

This study examines the performance of simulated annealing for minimizing a cost function defined on a two-dimensional discrete grid. The analysis focuses on the effects of temperature, problem size and the structure of the optimization landscape, before comparing annealing with random-walk and greedy-like search strategies.

The objective is not only to locate low-cost solutions, but to understand why the algorithm succeeds or fails under different conditions. The experiments and numerical results reported below are from the original study.

2 Move proposals and boundary handling

The modulo-n operation in the proposal method plays an important role in handling boundaries efficiently. At each state (i, j), a new index is randomly selected from $(i - 1) \bmod n$, $(i + 1) \bmod n$ and another from $(j - 1) \bmod n$, $(j + 1) \bmod n$. The resulting proposal is a diagonal step of size one in one of four directions.

The modulo operation allows the algorithm to wrap around the edges of the grid. For example, when $i = 0$, the index $(i - 1) \bmod n$ corresponds to $n - 1$. This produces a circular effect and simplifies boundary handling from both mathematical and implementation perspectives.

Without wraparound, states at the boundaries would have fewer valid neighbors. If proposals were sampled uniformly among those valid neighbors, forward and reverse proposal probabilities could differ. In that case the standard Metropolis acceptance rule would need a proposal-ratio correction, or the move mechanism would need to be modified to preserve symmetry. Boundary-specific conditions would also make the implementation more complex.

$$P(\text{accept}) = \min(1, \exp\left[-\beta\, \Delta C\right]) .$$

Here, ΔC is the proposed change in cost and $\beta = 1/\mathrm{T}$ is inverse temperature. This simple Metropolis rule applies to symmetric move proposals.

3 Temperature schedule and convergence

To analyze how performance changes with temperature, five pairs of inverse-temperature parameters (β_0, β_1) were considered: (0.001, 0.01), (0.01, 1), (1, 5), (5, 10), and (10, 20). The average cost reached at each of 50 annealing steps was recorded for a grid of size $n = 250$.

Within each temperature range, the algorithm behaves as expected. High initial temperature allows exploration, so average cost begins to decrease. As temperature falls and β increases, the algorithm becomes progressively less willing to leave low-cost states. In these experiments, lower β ranges generally performed better because they allowed the algorithm to explore different areas in the early stages, increasing the probability of reaching a minimum.

The analysis was repeated for $n = 2{,}000$. Although the optimizer rarely reached the global minimum, the general trend was similar. The average cost in these experiments did not fall

below approximately -55. Low β values, particularly around 0.01 to 0.1, gave relatively better results than the more restrictive schedules.

Values that are too small can also be ineffective: when the Metropolis acceptance probability is very high, the procedure behaves increasingly like a random walk and may leave low-cost regions after discovering them. Conversely, a greedy strategy with little initial exploration is highly sensitive to the starting point. Overall, simulated annealing can help locate minima, but success depends strongly on the temperature schedule.

4 Scaling with problem size

To examine convergence as the search space grows, the algorithm was run 20 times at each grid size $n \in 100, 200, 500, 1{,}000, 5{,}000$, keeping the other parameters constant. Two measures were collected: (i) the proportion of runs reaching a cost between -105 and -95, and (ii) the average annealing step at which this occurred, among successful runs.

Table 1: Original experimental results, based on 20 runs per grid size. No convergence time is reported when no run succeeded.

Grid size (n)	Success rate	Mean first-hit step
100	90%	0.06
200	65%	1.77
500	40%	13.63
1,000	15%	20.00
5,000	0%	—

For small problems, the algorithm often reached the minimum within the first few steps, suggesting that the success of these runs was not exclusively due to the annealing procedure: random initialization also allowed different parts of the relatively small search space to be visited.

As grid size increased, the success rate fell considerably and the number of annealing steps needed to reach a minimum increased. The low-cost regions did not grow proportionally with n; consequently, they occupied an increasingly small fraction of the full search space. Finding them within a limited computational budget therefore became much less likely.

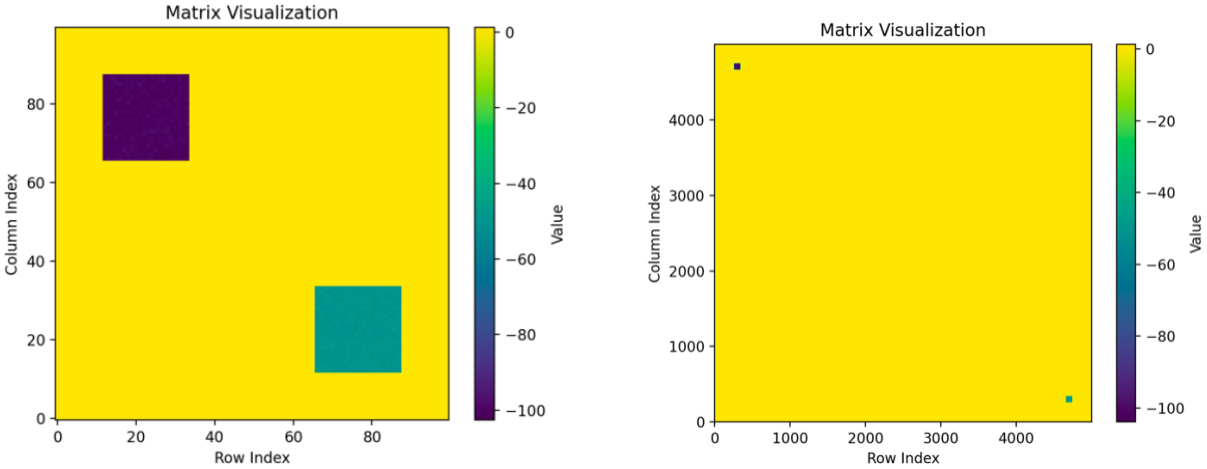


Figure 1: Original landscape plots for smaller and larger grids. The minimum regions occupy a substantially smaller fraction of the larger domain.

The results suggest that the local simulated annealing approach is more effective for small and medium-sized grids than for very large ones under the tested parameters. Performance might be improved by adjusting the number of Metropolis steps and the exploration schedule.

5 Move acceptance and the search landscape

The mean acceptance rate was recorded at each annealing step across 20 runs, first for $n = 250$ and then for $n = 1{,}000$. At the beginning, β is low and proposed moves are frequently accepted, even when they increase the cost. This supports exploration across the surface.

As annealing progresses, β rises and moves that worsen the cost are less likely to be accepted. The algorithm may also be inside a relatively low-cost region, where neighboring proposals are less likely to improve the objective. The acceptance rate consequently decreases toward zero.

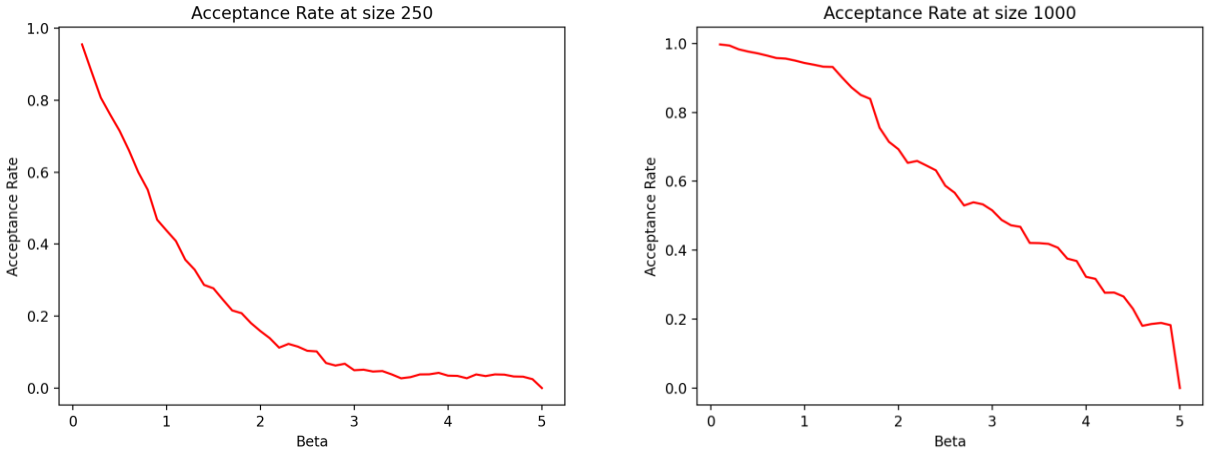


Figure 2: Original mean acceptance-rate curves for grid sizes n = 250 (left) and n = 1,000 (right).

At $n = 250$, the average acceptance rate declined rapidly. At $n = 1{,}000$, the decrease was slower and acceptance remained higher during the later stages. This is consistent with unsuccessful trajectories still moving through relatively flat, higher-cost areas, where neighboring moves change the cost only slightly.

Acceptance rates therefore provide insight into the region of the landscape being explored, but they are not, on their own, proof that the global minimum has been found. They need to be interpreted together with the objective values and the rate of successful runs.

6 Geometry of the objective function

The three-dimensional representation shows a nearly flat, non-monotonic surface, with costs near zero for most points (x, y). Two localized low-cost regions stand out: a deeper basin near -100 and a shallower basin near -50. The spatial extent of these regions does not increase proportionally with the size of the problem.

This geometry explains several observations. When the search starts far from a basin, the almost-flat background offers little information about where the minimum lies. High temperatures permit broader exploration. Once a minimum is reached, lower temperatures reduce the likelihood of climbing out of it, allowing further local refinement.

The shallow basin can also trap the algorithm. Escaping it requires accepting a substantial increase in cost, which becomes increasingly unlikely late in the annealing schedule. If the algorithm settles there instead of reaching the deeper region, it returns a suboptimal solution.

7 Comparison with random and greedy-like search

To evaluate whether simulated annealing was an appropriate method for this landscape, the original analysis compared three inverse-temperature regimes at grid sizes $n = 100$, 500 and 1,000:

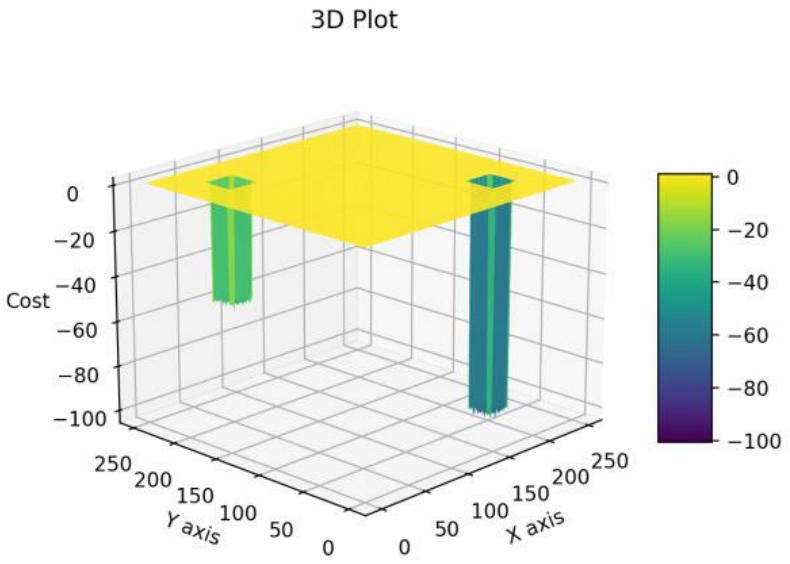


Figure 3: Three-dimensional visualization of the original objective function, showing the flat background and two localized minima.

- **Near-random:** β from 10^{-7} to 10^{-5}, with acceptance behavior close to a random walk.
- **Annealing:** β from 0.01 to 2, using the temperature range selected from earlier experiments.
- **Near-greedy:** β from 1,000 to 10,000, accepting very few worsening moves.

Table 2: Historical success rates from the original comparison. The random and greedy conditions are approximated through extreme inverse-temperature schedules.

Temperature regime	n = 100	n = 500	n = 1,000
Near-random	100%	75%	25%
Simulated annealing	95%	45%	15%
Near-greedy	10%	0%	0%

The comparison confirms that algorithm performance varies considerably with problem size. The near-random regime performed better than the tested annealing schedule, especially for small problems where continued exploration could cover much of the surface. The near-greedy regime was particularly ineffective because the small regions of low cost were difficult to locate without exploration.

As the grid grows, even random-walk behavior becomes less efficient. Annealing can also be unsuccessful if temperature decreases before the algorithm has reached a promising region; once in the shallower basin, escaping becomes difficult. These findings apply to the specific synthetic landscape tested and do not imply that a random walk is generally superior to simulated annealing.

8 Possible improvements and conclusion

A more effective approach for this problem could combine the benefits of randomness with an adaptive annealing procedure. One possibility is to extend the initial phase of global exploration, sample many candidate states and retain the best starting configuration found. This would improve the chance of beginning near a low-cost region.

A second improvement would be to vary the distance of proposed moves. Instead of moving a fixed distance of one cell throughout the process, the algorithm could start with larger, problem-

size-dependent jumps and reduce them gradually as annealing proceeds. Larger early moves would help explore the grid, while shorter later moves would permit local refinement.

Together, these changes could allow the algorithm to visit a broader region of the search space without increasing the computational budget excessively, potentially improving its probability of finding a high-quality solution.

Overall, the study illustrates how strongly optimization performance depends on objective-function geometry, initialization and the balance between exploration and exploitation. The observations are specific to the landscape analyzed and should not be generalized to unrelated optimization problems without further testing.